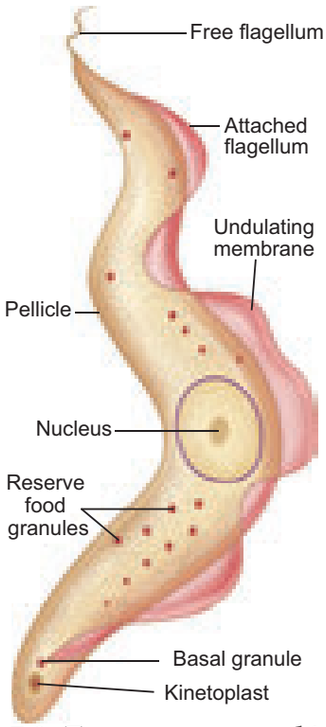


Fig. 7.2. *Trypanosoma gambiense*

blood vessels of the spleen. Symptoms of Kala azar are weight loss, anaemia, fever, enlargement of spleen and liver.

Malaria is caused by different types of *Plasmodium* species such as *P. vivax, P. ovale, P. malariae and P. falciparum* **(Table 7.3).** *Plasmodium* lives in the RBC of human in its mature condition it is called as **trophozoite**. It is transmited from one person to another by the bite of the infected female *Anopheles* mosquito.

Life cycle of *Plasmodium*

Plasmodium vivax is a digenic parasite, involving two hosts, man as the secondary host and female *Anopheles* mosquito as the primary host. The life cycle of *Plasmodium* involves three phases namely **schizogony, gamogony** and **sporogony (Fig. 7.3)**.

JHPWVF

The parasite first enters the human blood stream through the bite of an infected **female *Anopheles*** mosquito. As it feeds, the mosquito injects the saliva containing the **sporozoites**. The sporozoite within the blood stream immediately enters the hepatic cells of the liver. Further in the liver they undergo multiple asexual fission **(schizogony)** and produce **merozoites**. After being released from liver cells, the merozoites penetrate the RBC's.

Inside the RBC, the merozoite begins to develop as unicellular trophozoites. The trophozoite grows in size and a central vacuole develops pushing them to one side of cytoplasm and becomes the **signet ring stage**. The trophozoite nucleus then divides asexually to produce the **schizont.** The large schizont shows yellowish - brown pigmented granules called **Schuffners granules.** The schizont divides and produces mononucleated merozoites. Eventually the erythrocyte lyses, releasing the merozoites and haemozoin toxin into the blood stream to infect other erythrocytes. Lysis of red blood cells results in cycles of fever and other symptoms. This erythrocytic stage is cyclic and repeats itself approximately every **48 to 72 hours** or longer depending on the species of *Plasmodium* involved. The sudden release of merozoites triggers an attack on the RBCs. Occasionally, merozoites differentiate into **macrogametocytes** and **microgametocytes**. When these are ingested by a mosquito, they develop into male and female gametes respectively.

In the mosquito's gut, the infected erythrocytes lyse and male and female gametes fertilize to form a diploid zygote called **ookinete**. The ookinete migrates to the mosquito's gut wall and develop into an **oocyte**. The oocyte undergoes meiosis by a process called **sporogony** to form **sporozoites**. These sporozoites migrate to the salivary glands of the mosquito. The cycle is now completed and when the mosquito bites another human host, the sporozoites are injected and the cycle begins a new.

The pathological changes caused by malaria, affects not only the erythrocytes but also the spleen and other visceral organs. Incubation period of **malaria** is about 12 days. The early symptoms of malaria are headache,

TN_GOVT_XII_Zoology_chapter7.indd 117 24-02-2020 4.02.34 PM